

Allyn Young on increasing returns

CHARLES P. BLITCH

The phenomenon of increasing returns has virtually no place in the current theory of competition. It is recognized that there are increasing returns to scale with larger sizes of plant, but in order to preserve the competitive model it is assumed these are exhausted at low levels of output. Some reference is made to "external economies" generated by industry growth, but these are restricted to such rare and nebulous categories that they are unimportant. Only in the theories of imperfect competition and pure monopoly do increasing returns play a part. In the former, the small size of the firm's market does not allow for full exploitation of scale economies, while in the latter it is their relative importance compared to the firm's market which confers monopoly status. Neither model concerns itself with "external economies." Thus, in mainstream theory, increasing returns are associated with the restricted output and higher price of monopoly.

Contemporary and historical industry studies stress the role of increasing returns in the expansion of industrial output at lower prices. Economic historians regard increasing returns as a major factor in the industrial development of Britain and the United States. This paradox between theory and reality has led many economists to call for a new framework or paradigm to bring the two closer together for the purposes of explanation, prediction, and policy.

The present position of increasing returns and "external economies" in economic theory is an outcome of the so-called "cost controversy" of the 1920s. One of the participants in that debate was Allyn A. Young who was disturbed by the trend of economists

The author is Professor of Economics, Old Dominion University.

in pursuing logical rigor at the expense of analysis of modern economic problems. In a paper, "Increasing Returns and Economic Progress" (1928), Young reinterpreted the concept of increasing returns by removing it from a particular industrial environment and extending it to the whole sector. In Young's view, competitive equilibrium methodology was not only inadequate but was a positively misleading vehicle for the examination of the returns phenomenon. He emphasized broad and dynamic aspects of productivity growth as a source of economic progress.

Young sketched a theory of long-run growth for a mature economy. Young's insights and conclusions have not been followed by other economists, with the signal exception of Nicholas Kaldor (now Lord Kaldor), a former student of Young's at the London School of Economics. The failure of economists to apply his ideas to current competitive theory was largely a consequence of a cascade of events which overwhelmed the profession: Young's untimely death shortly after the publication of his paper; the onset of the worldwide depression; the monopolistic competition and Keynesian "revolutions"; the outbreak of World War II and subsequent postwar problems. The neglect was also due to the nature of Young's analysis of increasing returns. Kaldor, in a recent formal lecture, referring to Young's article explained: "... it was so many years ahead of its time that the progress of economic thought has passed it by. . . . Economists ceased to take any notice of it long before they were able to grasp its full revolutionary implications. . . . It was partly also because its importance as a basic criticism of general equilibrium theory could not be appreciated at a time when that theory itself was not properly understood" (1972, p. 1243).

The present essay attempts to reconstruct and interpret Allyn Young's thoughts on increasing returns in an advanced industrial economy. The major argument is that he was endeavoring to persuade economists to change their "vision" of such an economy from one of static general equilibrium with perfect competition, perfect foresight, homogeneous commodities and resources, and constant returns to a revised version of the growth model as expounded by Adam Smith, modified by Alfred Marshall, and updated by Young. The first part briefly summarizes the background of Young's work, while the second portion sets forth the details of his theory of economic progress. The third section deals with

the broader and revolutionary implications of Young's paper.

The controversy over increasing returns

Alfred Marshall devised the analytical concept of "external economies" for two reasons. First, along with "internal economies" it provided a means whereby the gains from the specialization of labor and capital could be introduced into his static constructions. Marshall, like Adam Smith, regarded the division of labor as a prime source of increasing returns and economic progress. Second, and in particular, since increasing returns could accrue as "external economies," it seemed possible to reconcile increasing returns and competitive equilibrium. Marshall recognized the logical difficulties that increasing returns posed for his static partial equilibrium method and inserted several warnings to that effect in his *Principles*. Nevertheless, he attempted to encompass the phenomenon within the constraints of his approach. Marshall's analytical device of "external economies" and his solution to competitive equilibrium with increasing returns were virtually unchallenged for thirty years. However, in the 1920s there began an intellectual ferment in the United States and in Britain which demanded a consistent logical rigor in the older theories. The decade commenced with the publication of Frank H. Knight's *Risk, Uncertainty and Profit* (1921). In this book Knight not only defined the concept of perfect competition in rigorous terms, but he also argued that for the purposes of economic analysis, a sharp separation must be made between the "static" and the "dynamic" problems. The "static" problem referred to the allocation of a given stock of productive resources in accordance with known wants with fixed technology and methods of business organization. The "dynamic" situation is one in which one or more of the fundamental conditions of the static state are changed. It deals with the maintenance and progress of society. Knight regarded the Marshallian concept of "external economies" and increasing returns as falling within the "dynamic" area. Thus, it was logically inconsistent to use it, as Marshall had, in static price theory.

Knight's doctoral thesis, on which the book was based, was completed under the supervision of Allyn Young, then a professor of economics at Cornell University. Young also reviewed the manuscript of the book prior to publication. Knight and Young

disagreed on the character, effects, and use of increasing returns in competitive theory. Their differences, the subject of a long and amiable discussion, are reflected in the various letters and manuscript notes exchanged by the two during the decade. On a page of the typed manuscript of this book, Knight had written:

It is usually assumed that cost may either increase, remain constant or decrease as supply is increased. . . . But examination seems to show that under conditions necessary to perfect competition, cost must always increase as supply increases. If there is to be competition, conditions must be such that an establishment of a relatively small size in comparison with the industry as a whole is more efficient than a large one; otherwise monopoly will result. New supply will then come through an increase in the number of similar establishments, . . . (1972, typed mss. #1, p. 36)

Young's handwritten marginal comments show that he did not think the clear distinction between the "static" and the "dynamic" areas was so essential.

"External economies" of a certain sort will be *realized* if the increased supply is a response to an increased demand. . . . The point is that certain economies are *possible only with large demand*. . . . But I admit that the economies of "large-scale production" are really economies of large-scale demand. . . . You miss the point, I fear, of Marshall's "external economies." They are the economies (in general) of greater specialization and div. of labor. They come with an *increase of output* in the industry. . . . An increased output means more *plants*, of course, but the important thing is that they are *not* "similar establishments" but, in general, more highly specialized establishments. (Ibid.)

These statements indicate the respective positions of the two disputants. Their views were to remain largely unchanged as the decade passed. In a journal article discussing the notion of social cost, Knight's opinion of "external economies" was publicly stated:

. . . the doctrine of "external economies," which surely rests upon a misconception. Economies may be "external" to a particular establishment or technical production unit, but they are not external to the industry if they effect its efficiency. External economies in one business unit are internal economies in some other, within the industry. Any branch or stage in the creation of a product which offers continuously a chance of technical economies with increase in the scale of operations must eventually . . . in monopoly. (1924, p. 172)

In a rejoinder to Frank Graham on his article, Knight made his feel-

ings about the concept of “external economies” in competitive analysis very clear. “In spite of the weight of authority which may be cited for such economies,” he said, “I have never succeeded in picturing them in my mind, or to find any convincing reason to believe they exist, . . . Until a plausible example is brought forward, the category of decreasing cost under stable competition remains an ‘empty economic box’” (1925, pp. 332-33).

Allyn Young, during the decade, was a member of the Advisory Committee to the Bureau of Census, a member of the Board of Directors of the National Bureau of Economic Research, and was appointed to the Economic Advisory Committee of the President’s Conference on Unemployment. These various organizations commissioned numerous reports, studies, and monographs which dealt with the different sectors of the American economy for the period 1921 to 1929. Young felt that the professional preoccupation with the logical rigor and refinements of static equilibrium theory was at wide variance with what the mounting empirical evidence showed. The data reflected an economy undergoing rapid change and growth in productivity in manufacturing, industrial structure, goods and services, and capital investment. Static equilibrium theory did not allow for growth and change, so Young’s dissatisfaction with such analysis grew with the passage of time. As early as 1922, in a letter to Frank Knight, he expressed his doubts about general equilibrium: “I have yet to see that the method of general equilibrium gives us anything at all that *gets us anywhere*” (Knight, 1972, AAY folder, 12/9/22). By the latter part of the decade, Young was considering abandoning equilibrium analysis as a way to examine increasing returns. He communicated this to Knight in a letter. Knight had published an article in which he criticized the use of the negatively sloped long-run supply curve resulting from “pure external economies.” Young replied: “Where I don’t follow you, of course, is in respect of increasing returns. The reason may be that increasing returns do not exhibit themselves adequately when approached from the point of *equilibrium* price-theory” (Knight, 1972, AAY folder, 8/11/28).

A contemporaneous discussion of the theoretical problems associated with “external economies,” increasing returns, and competitive equilibrium had developed in Britain. It involved such prominent names in British economics as J. H. Clapham, A. C. Pigou, D. H. Robertson, and Gerald Shove. It was brought to a climax

with the publication of Piero Sraffa's famous criticisms of Marshallian economics (1926).

Sraffa also attacked "external economies" on the basis of logical inconsistency. He pointed to Marshall's admission in *Industry and Trade* that "external economies" could seldom be allocated to a single industry but are often attached to groups of correlated industries. In this case, "external economies" are incompatible with partial equilibrium methods. To save the concept, said Sraffa, it was necessary to confine such economies to those that were "external" to the firm but "internal" to the industry. Sraffa, agreeing with Knight, thought these types of economies were practically nonexistent. To treat decreasing costs under conditions of partial equilibrium analysis, stated Sraffa, it was necessary to drop the assumption of perfect competition and use monopoly theory.

The reasoning and conclusions of Sraffa's article, reinforced by Knight's earlier work, shook the foundations of the Marshallian edifice. Rebuttals, confirmations, and modifications in the form of classroom lectures, public addresses, and scholarly papers began to accumulate. Allyn Young arrived in London in the midst of this theoretical turmoil to assume the chair of political economy in the London School of Economics. He did not agree with Sraffa, of course, that the solution to the problem of increasing returns was to turn to the theory of monopoly. Young was reworking his view on "external economies" and competition. He had concluded that in order to study this relationship it was necessary to abandon static methods. This realization he communicated to Knight: ". . . the static view does not interest me very much, because; if it is *rigorously* adhered to, almost everything worth saying about it can be put onto a very few pages. We *have* to depart from it somehow. The only question is just how" (Knight, 1972, AAY folder, 10/6/28).

Shortly after arriving in Britain, Allyn Young was elected president of Section F (Economics and Statistics) of the British Association for the Advancement of Science—the first American to be so honored. This gave him the opportunity of presenting his thoughts on the question of increasing returns and competition before a large part of the British economic establishment. On September 10, 1928, he delivered his presidential address at Glasgow University—an appropriate place, for the paper was basically an extension and refinement of Adam Smith's theory of economic progress.

Allyn Young's theory of economic progress

Adam Smith's theorem that the division of labor is limited by the extent of the market was the central theme of Young's address. "That theorem," Young said, "I have always thought, is one of the most illuminating and fruitful generalizations which can be found anywhere in the whole literature of economics" (1928, p. 529). And it was on the basis of this simple idea that he constructed an elaborate explanation of economic expansion in an industrial economy. Like Smith, Young believed that agriculture did not offer as many opportunities for the division of labor as did manufacturing. Then, too, the growth of that area was limited by the inelastic supply of land. The data on the U.S. economy for the decade supported Young's belief for it showed a startling advance in industrial productivity with only moderate gains in that of the farm sector.

The division of labor in its modern form, according to Young, is more far-reaching than the specialization of activities in a single craft illustrated by Adam Smith's pin-making example. It manifests itself in two related aspects: the growth of indirect or roundabout methods of production and specialization among industries. Young stressed the fact that in advanced countries production methods are capitalistic; that an unfailing characteristic of capitalistic production is roundaboutness; and roundaboutness involves the breaking up of a complicated process into a succession of simpler processes, many of which lend themselves to the use of machinery. Young made the point that these economies depend upon the extent of the market. With a small output it is more economical to rely upon methods making a relatively larger use of directly applied labor and a relatively smaller use of indirectly applied labor. Kaldor has argued that this means that the techniques chosen (capital-labor ratio) are a function not of relative factor endowments or factor prices but of the level of production (1972, p. 1242).

The specialization of industries was seen by Young to be the most important source of increasing returns: "the progressive division and specialization of industries is an essential part of the process by which increasing returns are realized" (1928, p. 439). Both empirical and historical evidence supported Young's argument that "industrial differentiation, has been and remains the type of change

characteristically associated with the growth of production" (p. 537). With economic advance there is continual structural change in the manufacturing region of the economy leading to greater specialization and interdependence. "External economies" are related not so much to the growth of a single industry but more to the volume of output of allied industries. For example, an increase in the demand for detergents may make the use of automatic packaging machines economical and thus reduce the cost of production not only of detergents but of the machines as well. The fall in the cost of the machines may make their use in the making of corn flakes and cocoa profitable and so on. "External economies" are associated with aggregate production and are the economies of roundabout or capitalistic methods of production.

Taking an overall view of industrial activity required a redefinition of the market. Young defined the market and its size as follows: "In an exclusive view, considering the market not as an outlet for the products of a particular industry, and therefore external to that industry, but as an outlet for goods in general, the size of the market is determined by the volume of production" (p. 533). Underlying this definition of the market is the Classical world of Say's Law. Goods are ultimately exchanged for goods, and productivity and income are measured in real terms. Young used this version of economic activity which abstracts from the use of money and the possible lack of effective demand, for "some aspects of important economic problems are seen more clearly when values are looked upon as though they were determined by direct comparisons of goods with goods without the intervention of money" (1929, p. 580). As long as sufficient aggregate demand can be assumed, Young provides an important insight into the supply-side aspects of economic progress. But his analysis is flawed unless one assumes active governmental policies to assure a full-employment level of aggregate demand.

The cornerstone of economic progress, said Young, is the creation and expansion of markets. He allotted this component a key part:

It is dangerous to assign any single factor the leading role in that continuing economic revolution which has taken the modern world so far away from the world of a few hundred years ago. But is there any other factor which has a better claim to that role than the persistent search for markets? No other hypothesis so well unites economic history and economic theory. (1928, p. 536)

Like Schumpeter, Young attributed much to the "entrepreneurial spirit" in economic change. "The businessman's mercantilistic emphasis upon markets," he wrote, "may have a sounder basis than the economist who thinks in terms of economic statics is prone to admit" (p. 537). One of the chief tasks of modern industry, Young observed, is finding markets, and the search for markets is not just a matter of multiplying sales but of adding to sales and profits by reducing costs. Young also suggests that "selling expenses" cannot be regarded as sheer economic waste for it depends upon their effect upon aggregate industrial product as opposed to their effects upon the profits of individual enterprises (p. 537).¹

Young argued that the division of labor, the major source of increasing returns, and the extent of the market were interdependent, but this could not be seen within the confines of partial analysis: "the mechanism of increasing returns is not to be discerned adequately by observing the effects of variation in the size of the individual firm or of a particular industry, . . . What is required is that industrial operations be seen as an interrelated whole" (p. 539). The extension of the division of labor required additional capital investment. Young treated investment in its cost-reducing sense, which is proper for the problem of growth and change. By increasing productivity per worker, it makes both higher wages and incomes possible; and by producing goods in more efficient and cheaper ways, it releases consumer income for expenditures on other goods and services.

Young protested the application of equilibrium methodology to what he saw as a process of change: "No analysis of forces making for economic equilibrium . . . will serve to illumine this field, for movements away from equilibrium, departures from previous trends are characteristic of it" (p. 528). He felt that the changes in production methods and industrial organization induced by the extension of the division of labor were not systematic enough to

¹ It is obvious from these remarks that Young is not considering competition in the sense of "perfect competition." Instead he is harking back to a version of competition used by Adam Smith. G. B. Richardson recently argued that Smith used two functions of competition. On the one hand, competition acted as an ordering and leveling device leading toward the establishment of market equilibrium. On the other hand, it stimulated market expansion by undermining that tendency toward equilibrium. It was competition in the form of free entry and the exploitation of additional opportunities for the division of labor (1975, pp. 350-51). It was this latter role of competition that both Smith and Young thought vastly more important.

be subjected to any type of equilibrium analysis, partial or general. "Every important advance in the organization of production," he wrote, "... alters the conditions of industrial activity and initiates responses elsewhere in the industrial structure which in turn have a further unsettling effect. Thus change becomes progressive and propagates itself in a cumulative way" (p. 533).

Allyn Young outlined a growth model for an advanced economy which incorporates his wider notion of increasing returns and employs an approach designed to take account of progress as a cumulative process. In his model Young made use of the reciprocal demand or offer curve which had been developed by Marshall and extended by Edgeworth to explain value in international trade. Among all neoclassical techniques it came closest to giving a general picture of exchange. Along such a curve goods exchanged for goods. It shows the quantity of any one good, say steel, which steel producers would supply to producers of another good, say textiles, for different amounts of textiles at various prices of textiles in terms of steel. Thus the steel producers' offer curve is simultaneously their supply curve for steel as well as their demand curve (of a kind) for textiles. The curve subsumes demand and supply conditions within each industry. In this pure barter situation, demand and supply are interdependent, as the supply of steel represents a demand for textiles just as the supply of textiles is a demand for steel. Edgeworth pointed out that "movements along a supply and demand curve of international trade should be regarded as attended with rearrangements of internal trade; as movements of the hand of a clock corresponds to considerable unseen movements of machinery" (1925, p. 32). He also noted that along such a curve, the cost of production and the marginal utility of both commodities as well as prices are released from the pound of *ceteris paribus* (pp. 32-33).

To establish the necessary and sufficient conditions for a cumulative process, Allyn Young postulated that the two goods are produced competitively under conditions of increasing returns, and that the demand for each commodity was "elastic, in the special sense that a small increase in its supply will be attended by an increase in the amounts of other commodities which can be had in exchange for it" (p. 534). In Young's opinion, the functioning of Say's Law required that the increased supply be produced under conditions of increasing returns and that the demand for commodities be elastic. The presence of increasing returns is necessary so

that an increase in supply results in lower costs, and the role of competition is to diffuse lower costs in lower prices. The elastic demand is essential so that the lower exchange value of one commodity in terms of another results in a larger demand for the other good. Young believed that these assumptions of increasing returns and elastic demand were sufficient for a self-generating process:

Under such circumstances an increase in the supply of one commodity is an increase in the demand for other commodities, and it must be supposed that every increase in demand will evoke an increase in supply. . . . Even with a stationary population and in the absence of new discoveries in pure and applied science there are no limits to the process of expansion except the limits beyond which demand is not elastic and returns do not increase. (p. 534)

Economic expansion is a process which requires time, and the time path of industrial progress is neither smooth nor easy. Young saw two major obstacles. First, human beings are resistant to change, and the process requires that new trades be learned and new habits acquired. There has to be a geographical redistribution of population, and established communities have to be broken up. Second, the accumulation of capital takes time even though it is a matter of turning some part of an increasing product into various forms of capital. In some fields, moreover, "The next important step forward is initially costly and cannot be taken until a certain quantum of prospective advantages has accumulated" (p. 535). In addition the demand for some products is inelastic or with increasing supply becomes so. There are natural scarcities or inelastic supply which blocks the securing of "external economies" in some products. There are, however, reinforcing factors in the division of labor among industries which are independent of changes in production techniques. It allows a higher degree of specialization of management; it lends itself to a better geographical distribution of industrial operations in terms of nearness to sources of supply of raw materials and/or cheaper power, to complementary industries, or to large population centers (pp. 535-36).

Young closes his remarks on increasing returns with a brief but provocative summary of his argument: "the division of labor depends upon the extent of the market, but the extent of the market depends upon the division of labor. In this circumstance lies the possibility of economic progress" (p. 539).

Conclusions

Allyn Young's address was a challenge to the methodology and purpose of neoclassical economics. "External economies" were not seen as analytical curiosities of doubtful significance but were considered as the prime source of increasing returns and economic progress. The essence of economic progress is growth and change. These elements cannot be encompassed within the constraints of equilibrium models of the static variety. Development is sequential and cumulative in nature, and each step forward is in large degree determined by the character of prior sequences. The whole notion of equilibrium itself is not useful for analyzing increasing returns, for each change begets another with reverberations throughout the structure of the economy. Given these destabilizing effects of increasing returns, a unique position of general equilibrium cannot be postulated.

Kaldor has argued that once the concept of a determinant general equilibrium position is denied, then the idea of a socially correct allocation of resources has no meaning (1972, p. 1245). An optimal distribution of labor, for example, among various employments cannot be described in a situation where every change in the organization of production redeploys some labor into new occupations with higher productivity. An optimum combination of factors for production based upon relative factor proportions or factor prices becomes nebulous if the heterogeneity of labor due to nature and training and of capital resulting from design and manufacture is admitted. Then the "best" choice depends entirely upon the size of the market for the output.

Finally, Allyn Young's model incorporating increasing returns eschewed the assumption that changes in cost and demand conditions were exogeneous. The entrepreneur is a principal agent of "change and progress," so the forces making for change were generated within the economic system. Taking a cue from economic history, Young argued that it was "the search for markets" that was the propelling factor. Market expansion allowed for extension of known aspects of the division of labor in production which would be too costly otherwise. The enlargement of any one market generating increasing returns would create "external economies" in other markets. It is a cumulative advance with action and reaction taking place *via* the division of labor and increases in market size.

What are the consequences of the Youngian frame of reference for contemporary economics? Does it mean "the threatened wreckage" of the greater part of economic theory? The main effect would be the removal of neoclassical value theory, with its attendant focus on the allocation of a given quantity of resources via the functioning of perfectly competitive markets, from center stage in economics. Prices would no longer be indices of the relative scarcity of commodities and resources. The principal function of prices would be to act as signals and incentives and to transmit the benefits of economic progress to consumers and resource owners alike. The education and continuing study of professional economists would have to undergo revision. "Pure theorists," skilled only in deductive logic and mathematics, would have to supplement their knowledge with economic history and statistics. The jettisoning of the current paradigm from its commanding position will also mean a rewriting of textbooks, professional publications, and a new approach to economic policy. Therefore, the acceptance of Young's analysis will incur costs to the economics profession. The cardinal benefit, however, is that economic science can resume its more important role—that of describing and predicting how an economy actually works—thereby laying the basis for intelligent and fruitful economic policy.

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